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Exploring the relationships between web accessibility, web traffic, and university rankings: a case study of Jordanian universities

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Abstract: Most statistics state that disabled people constitute approximately 20% of the world population; therefore, webmasters should consider this part of the world population when they design and implement their websites by following the web content accessibility guidelines. This paper aims to explore the relationships between web accessibility, web traffic (web), and university rankings using the metrics of 27 Jordanian university websites as a case study. Objective evaluations on the web accessibility, web traffic (website popularity), and university rankings of 27 Jordanian university websites are conducted using a number of online tools. An extensive analysis of the tool outputs is presented, and the relationships of these tool outputs are explored. This study represents a longitudinal overview of the accessibility of Jordanian university websites and re-explores the different relationships between web accessibility, web traffic, and university rankings, as well as the differences between the accessibility of private and public Jordanian universities. This paper explores the relationships between three parameters: accessibility, web traffic, and university rankings. Therefore, its scope is larger than earlier publications.

Keywords: university websites; accessibility guidelines WCAG 1.0 and WCAG 2.0; web accessibility; web traffic; website popularity; university rankings; automated evaluation; Jordan.

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Biographical notes: Mohammed N. Al-Kabi received his PhD in Mathematics from the University of Lodz/Poland (2001), Master's degree in Computer Science from the University of Baghdad/Iraq (1989), and Bachelor degree in Statistics from the University of Baghdad/Iraq (1981). He is an Assistant Professor in the Computer Science Department, Faculty of IT at Zarqa University. Prior to joining Zarqa University, he worked at Yarmouk University in Jordan, Nahrain University for 11 years and Mustanserya University in Iraq for six years. He also worked as a part-time Lecturer at Jordan University of Science and Technology (JUST), Princess Sumaya University for Technology (PSUT) and Sunderland University. His research interests include sentiment analysis and opinion mining, big data, information retrieval, web search engines, data mining, social media, and natural language processing. He is the author of more than 94 peer-reviewed articles on these topics. His teaching interests focus on information retrieval, big data, web programming, data mining and DBMS (ORACLE and MS Access).

1 Introduction

The World Wide Web (WWW) has witnessed the development of many new technologies to present information since its onset. The concept of WWW is different from the concept of cloud computing, where the services in cloud computing include providing shared computing resources across the internet to different machines (Do and Hussain, 2013). The scope of this study does not include any requirements validation of cloud systems (Moscato, 2015). The evolution of the WWW should involve providing web accessibility to disabled people. In addition to adopting web content accessibility guidelines (WCAG) in their websites, therefore, webmasters should ensure that their websites could accommodate new devices (smartphones, tablets, and computers regardless of their screen size) to benefit disabled internet users who use their websites. These websites are becoming more complex; hence, their compliance with WCAG is becoming more difficult. The first version of WCAG (WCAG 1.0) was released in 1999 and consists of 14 guidelines; each guideline is subdivided into checkpoints. The total number of these checkpoints is 65, and each checkpoint has three

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priority levels (Priority 1, Level A conformance; Priority 2, Level AA conformance; and Priority 3, Level AAA conformance). The second version of WCAG (WCAG 2.0) was released in 2008, and it consists of 12 guidelines and success criteria organised around four principles. Each success criterion describes specifically what must be achieved to conform to the guideline to which it belongs. The total number of success criteria in this version is 61. Websites that comply with these necessary guidelines ensure an accessible web environment. Manual evaluation is thought to be the most suitable choice to assess accessibility, but it is no more a feasible choice because the size of many websites is becoming unmanageable. In this approach, a specialised team is recruited to conduct a manual evaluation. Funds need to be allocated to the team, and waiting for their results takes time. Automatic accessibility evaluation tools are the most desirable choice because many of these tools are free and do not need a long waiting period for the results. These automatic online accessibility evaluation tools can help to correct accessibility errors. They also widely differ in their capabilities. These errors are different from those faults of different software systems (Munawar and Ward, 2011). However, many checkpoints can be manually tested only; hence, using these tools will not guarantee web accessibility. Evaluations with disabled internet users should be conducted to identify specific problems. Webmasters who plan to build accessible websites should use these evaluation tools during the web development life cycle.

WebAIM's web accessibility versatile evaluator (WAVE) (http://wave.webaim.org/) uses an application program interface and is one of the most satisfactory free online tools for testing whether web pages are accessible to blind people. The WAVE tool was originally developed and sponsored by the Temple University Institute on Disabilities in Pennsylvania. The WAVE tool analyses the web page under consideration automatically and embeds icons and indicators. The WAVE tool also generates graphical and readable output for visually capable evaluators. WAVE enables web developers to identify most accessibility problems (accessibility violations) on their web pages. Red icons indicate accessibility errors, yellow icons indicate alerts or possible accessibility error, green icons indicate features, blue icons indicate structural or semantic elements, and purple icons indicate the use of HTML 5 structural elements or ARIA roles, states, or properties. Black icons indicate colour contrast errors according to WCAG guidelines. The WAVE tool is a quick and efficient way to identify many accessibility problems and features of the websites, and is thus used in the present study. WebAIM's WAVE accessibility reporting page does not provide its evaluation report in terms of 'pass' or 'fail'. Two additional free online tools are used to assess accessibility: AccessMonitor (http://www.acessibilidade.gov.pt/accessmonitor/) and the European Internet Inclusion Initiative (EIII) Page Checker (http://checkers.eiii.eu/). Accessible websites enable normal and disabled people to visit them; consequently, the number of visitors (web traffic/website popularity) will increase, which may have a positive effect on the rank of university websites.

The higher education sector plays an essential role in building a generation of thinkers in different fields of science, technology, and humanities. Therefore, many countries, including developing countries, realise the role that science can play in driving economic development. National governments should invest in their science base to secure economic prosperity even in a period of economic turbulence. Hence, many national governments have presented a number of reforms in education and invested in infrastructure and projects designed to create knowledge-based economies. Education reforms take place when governments recognise that technology and innovation are key elements to achieve long-term economic and social development. Thus, expenditure on research and development is increased significantly by governments that have realised the significance of this type of investment in the past few years. In most cases, increased investment in research and development leads to increased university rankings. A number of magazines, universities, newspapers, websites, and governments conduct the ranking assessment of different higher education establishments worldwide. Therefore, many global rankings of colleges and universities exist; these rankings are different because of the difference in the algorithms that were used for ranking. In the present study, Jordan is selected because it lies in the middle of the list of Arab countries by population (Wikipedia 2016). In addition, according to Elgohary (2008), Jordanian universities constitute 40% of the top 10 Arab universities. To identify the ranks of 27 Jordanian universities, the present study is based on the Webometrics Ranking of World Universities published by Cybermetrics Lab because it is the only comprehensive university ranking that includes the rank of the 27 Jordanian universities.

Some specialists define web traffic as the number of visitors to a website; others define it as the amount of data sent and received by internet users to a website, excluding the traffic generated by web crawlers. Generally, the number of visitors to a website is not distributed equally on its web pages. Hence, the web traffic to each website is monitored by the website owner to identify popular web pages and links. Many free tools can be used to monitor website traffic, such as Google Analytics and Alexa.com. Alexa.com is used in this study to identify the ranks of 27 Jordanian university websites on the national and world levels; the national rank is used to identify its relationships to accessibility and university rankings.

This paper aims to explore the homepages of 27 Jordanian websites to determine their accessibility barriers (accessibility violations) and rank these websites according to their web traffic, web accessibility scores, and university rankings.

Websites that comply with WCAG are called accessible websites to people with disabilities. This study also aims to rediscover whether any relationship exists between

accessibility, web traffic, and university rankings. Moreover, this study tests different hypotheses about the relationships between accessibility, web traffic, and university rankings. Although prior studies such as Lilly et al. (1999), Schmetzke (2001) and Kane et al. (2007) show no significant correlation between the ranks of the universities and their accessibilities, we conduct this study to rediscover these relationships, and found no significant correlations between the ranks of the universities and their accessibilities. Kendall's tau and Spearman's (rho) rank correlation coefficients are used to determine the relationships between accessibility, web traffic, and university rankings. The scope of this paper is larger than scope or earlier studies, wince it explores the relationships between three parameters: accessibility, web traffic, and university rankings.

The remaining parts of this paper are organised as follows: Section 2 presents a summary of a number of studies related to exploring the relationships between web accessibility, web traffic, and university rankings. Section 3 presents the methodology adopted in this study. Section 4 discusses the statistical methods. Section 5 presents the conclusions and future plans to improve this study.

2 Literature review

The relationship between accessibility and popularity of websites is presented by a number of studies, one of which is conducted by Lilly et al. (1999). Those authors include the library home web pages of Yahoo!'s list of the '100 Most Wired Colleges' in their study; they also use Bobby tool to analyse the accessibility of the web pages' understudy. The Bobby tool identifies 40 library homepages as accessible, and the study concludes that no relationship exists between accessibility and popularity. Our new study concludes that a negative relationship is present between accessibility and web traffic. In addition, Schmetzke (2001) uses the US News and World Report to determine the library home web pages of the top 24 schools of library and information science in the U.S. He discovered that 4 out of the 24 library websites are 100% accessible and that no relationship exists between accessibility and popularity.

Hackett et al. (2004) use the Internet Archive's Wayback Machine to select a random sample of websites from 1997 to 2002. Their sample consists of 240 websites that are distributed equally within those six years. They aim to study the effects of technology on accessibility, and they conclude that the adoption of new technologies in web pages increase their complexities, thereby increasing accessibility barriers.

The accessibilities of university websites are studied by many authors because these institutions should offer knowledge indiscriminately to all people whether they are disabled or not. One of these studies is conducted by Kane et al. (2007), which included a multi-method analysis of the homepages of the top 100 universities in the world; Kane et al. (2007) consider the homepage as a representative of each university's website, which is a debatable issue. These homepages are analysed for their compliance with accessibility standards, choice of language, image accessibility, text-only content, and quality of web accessibility statements. The effects of language, geographic location, public funding status, and university ranking on university's level of web accessibility are also studied in their study. They conclude that a significant geographical variation exists in the website accessibility of these 100 universities in addition to the accessibility errors; many of those websites lack accessibility policies or documentation. Few homepages approach full accessibility, specifically those in English-speaking countries, such as the USA, UK, Canada, and Australia. The accessibility of the homepage of each public Portuguese university is examined by Espadinha et al. (2011); they used two recommended online tools, namely eXaminator and HERA, by the Portuguese government to measure the conformance of the websites under study with WCAG checkpoints and detect errors. They conclude that only 12.5% of these websites are accessible.

Comeaux et al. (2007) investigate the websites of 56 American Library Association-accredited library schools. They use manual and automatic evaluation of these websites. Bobby 3.1.1 tool is used to evaluate the accessibilities of these sites automatically. Their results indicate an improvement in accessibility, but the used tool could not detect any Bobby-detectable errors in 50%–60% of the 56 websites under study. They conclude that Canadian websites are more accessible than US websites, and a relationship exists between popularity and accessibility. They also test the accessibility of each of the 56 websites of North American academic libraries (Comeaux et al., 2013). Furthermore, they test the relationship between table- and CSS-based designs and accessibility, and conclude that a larger sample should be used to determine the effect of these designs on accessibility.

Two studies by Lopes and Carriço (2010) and Lopes et al. (2010) show high variability on the accessibility level of different web pages, and the low percentage of these web pages gains high accessibility levels. Furthermore, these two studies conclude that a correlation exists between accessibility and complexity.

Kaur et al. (2014) investigate the accessibilities of bank websites in India and find that all bank websites in India have accessibility barriers; their accessibility evaluation reveals no difference among the (private and public) bank websites in India. They also discover a positive correlation between complexity and importance, and a negative significant correlation between complexity and popularity. In addition, they conclude that no significant relation exists between accessibility from one side and complexity, importance, and popularity from the other side.

Buenadicha et al. (2001) propose a new web assessment index (WAI); their index is based on four key quality factors (accessibility, speed, navigability, and website content) that are essential to determine the quality of a website. They use their index to evaluate the websites of all

Spanish universities and determine each of their score (based on a total of 100 points). Buenadicha et al. (2001) present a high correlation between site popularity and university size (number of students). A similar study by Gonzalez et al. (2004) proposes the WAI, which is based on the same four key factors presented by Buenadicha et al. (2001); these authors show a high correlation value between site popularity and firm size. Furthermore, Zeng (2004) concludes that a significant relationship exists between WAB score and Alexa traffic ranking in his Ph.D. thesis; however, no statistically significant relationship is observed between WAB and PageRank scores.

Typically, the empirical analysis of the relationship between website ranking and accessibility score involves a bivariate correlation analysis; such analysis is conducted by Kamoun et al. (2014) using 21 websites that are affiliated with the Dubai e-government. They conclude that a weak correlation exists between e-government website ranking and their web accessibility scores.

The relationship between accessibility and complexity is also explored by Hackett et al. (2004); they discover that governmental websites remain accessible and that using new technologies within web pages lead to an increased complexity, thereby increasing the number of accessibility barriers.

Many studies explore the correlation between the accessibility scores of (URLs/websites) and other metrics, such as web rank and complexity. Freire et al. (2008) study the correlation between different web accessibility metrics and also present a good survey of recent progress in this area of study.

Accessibility is considered a subset of usability by some experts in this field, whereas others consider usability a subset of accessibility. Therefore, a number of studies were conducted to show the relation between accessibility and usability. Petrie et al. (2007) show an overlap in the problems that face both disabled and non-disabled people; however, they fail to find any relationship between the ratings of the participants, researchers, and usability guidelines regardless of whether they are disabled or not. They conclude that accessibility problems are not a complete subset of usability problems and that usability problems are not a complete subset of accessibility problems. Sullivan et al. (2000) also conclude a weak correlation between content accessibility and overall usability.

3 Methodology

This section is divided into three subsections. The first subsection presents the selection of websites used in this study, the second subsection shows the measurements and tools used in this study, and the third subsection shows the statistical methods adopted in this research.

3.1 Selection of websites

First, a sample of websites is collected as a case study. The websites of all 27 accredited universities from the Jordanian Higher Education Accreditation Commission (http://www.heac.org.jo/) are selected for this study because of our extensive experience as a faculty member in some of these universities.

Accreditation means implicitly improving the quality of education to all people in the society regardless of whether they are disabled or not. Therefore, universities should consider accessibility and try to offer accessible websites to their visitors. Currently, accredited universities focus on their ranks on the national and international levels, and conduct high-quality research to improve their ranks. Supposedly, accessible websites attract more visitors relative to inaccessible ones. Hence, accessibility, web traffic, and university rankings have to be correlated with one another. Therefore, this study explores the relationships between accessibility and web traffic, accessibility and university rankings, and web traffic and university rankings.

3.2 Evaluation of web accessibility, web traffic, and university rankings

A number of free online tools are used in this study. The first tool is called WAVE and is used to identify errors (WCAG Violations), alerts, features, structural elements, HTML 5, ARIA, and contrast errors. The second and third tools are AccessMonitor and EIII, which are used to evaluate accessibility. Alexa.com is the fourth tool used to measure the web traffic of different web pages. Webometrics ranking of world universities is used to identify the university rankings of the 27 accredited Jordanian universities under study.

3.3 Statistical methods

The non-parametric statistical analysis is used to analyse the collected data about 27 Jordanian universities. This study uses Kendall's tau-b and Spearman's rho rank correlations to determine the degree of concordance between two columns of ranked data. The coefficients of these two correlations will decrease as the number of inversions in the two ranks increases. The coefficient range of these two correlations is between −1.0 and 1.0. Generally, Kendall's tau_b obtains better estimations (probabilities) on population parameters relative to Spearman's rho correlation, specifically for a small sample size. Spearman's rho correlation obtains more accurate results than Kendall's tau-b in the case of large discrepancies between two columns of ranked data.

4 Experiments and results

WebAIM's WAVE tool (http://wave.webaim.org/) failed to process the homepages of Al Isra Private University in Amman and Jadara University on August 28, 2015, due to a shutdown in their web servers, but this tool processed the homepages of the 25 other Jordanian university websites and obtained the results presented in Table 1. The results of the WAVE tool related to the homepages of Jordanian universities are presented in Table 1 and sorted in ascending order based on their sum of all errors and alerts (web accessibility violations) as follows: American University of Madaba: 4; Yarmouk University: 13; Jordan University of Science and Technology: 24; Al Hussein bin Talal University: 34; Amman Arab University: 60; Mutah University: 64; Arab Academy for Banking and Financial Sciences: 65; University of Jordan: 72; Ajloun National University: 74; and Al Al-Bayt University: 74. One may deduce from Table 1 that the homepage of American University of Madaba website is more accessible than the 24 other homepages of Jordanian universities because it has the fewest errors and alerts. Some may also argue that the results of the WAVE tool clearly show that the homepage of American University of Madaba Website lacks the features, structural elements, and HTML 5/ARIA tags used by the 24 other homepages of Jordanian universities; in such case, judgments should consider these features, which is an appropriate approach. Web accessibility tests check compliance with WCAG. WebAIM's WAVE tool does not show whether the web page understudy achieves a 'pass' or 'fail' on the accessibility test; therefore, the sum of errors and alerts is the most desirable indicator to show compliance with accessibility guidelines. Table 1 shows the rank of each university in the list on the basis of the sum of errors, alerts, and contrast errors in their homepages and not in the entire website. Therefore, Table 1 values are sorted in ascending order according to the sum of three values (error, alert, and contrast error numbers). The highlighted values in Table 1 represent either a maximum or a minimum value within a column. The minimum, maximum, and an average number of universities are below average (better half) of each of the eight columns of errors and alerts; they are presented at the bottom of Table 1.

Table 1 Summary of WAVE tool results ranked according to the sum of errors, alerts, and contrast errors (August 28, 2015 and March 1, 2016)

#	*University name*	*University URL*	*Errors*	*Alerts*	*Features*	*Structural elements*	*HTML 5 and ARIA*	*Contrast errors*	*Sum of errors, alerts, and contrast errors*	*Sum of features, structural elements, and HTML 5/ARIA*
1	American University of Madaba	http://aum.edu.jo/	3 3	1 1	0 0	0 0	0 0	0 0	4 0	0 0
2	Yarmouk University	http://www.yu.edu.jo/	3 6	8 10	24 38	76 82	0 0	2 2	13 18	100 120
3	Jordan University of Science and Technology	http://www.just.edu.jo/	1 ?	9 ?	11 ?	22 ?	0 ?	14 ?	24	33
4	Al Hussein bin Talal University	http://www.ahu.edu.jo/	15 21	2 3	19 21	27 29	1 11	17 16	34 40	47 61
5	Amman Arab University	http://www.aau.edu.jo/	11 ?	19 ?	15 ?	32 ?	9 ?	30 ?	60 ?	56 ?
6	Mutah University	http://www.mutah.edu.jo/	8 8	46 65	31 40	41 48	0 2	10 2	64 75	72 90
7	Arab Academy for Banking and Financial Sciences	http://www.aabfs.org/	34 ?	24 ?	9 ?	29 ?	0 ?	7 ?	65 ?	38
8	University of Jordan	http://www.ju.edu.jo/	32 ?	23 ?	39 ?	127 ?	0 ?	17 ?	72 ?	166 ?
9	Ajloun National University	http://www.anu.edu.jo/	16 ?	10 ?	27 ?	23 ?	4 ?	48 ?	74 ?	54 ?

Table 1 Summary of WAVE tool results ranked according to the sum of errors, alerts, and contrast errors (August 28, 2015 and March 1, 2016) (continued)

#	*University name*	*University URL*	*Errors*	*Alerts*	*Features*	*Structural elements*	*HTML 5 and ARIA*	*Contrast errors*	*Sum of errors, alerts, and contrast errors*	*Sum of features, structural elements, and HTML 5/ARIA*
10	Al Al-Bayt University	http://www.aabu.edu.jo/	7	44	8	12	3	23	74	23
			8	46	8	20	5	35	89	33
11	Jerash Private University	http://www.jpu.edu.jo/	33	13	3	28	0	31	77	31
			62	20	3	34	0	27	109	37
12	Tafila Technical	http://www.ttu.edu.jo/	29	27	21	37	0	27	83	58
			47	25	23	40	34	20	92	97
13	Applied Science	http://www.asu.edu.jo/	78	10	20	48	0	0	88	68
			67	36	22	51	0	0	103	73
14	Middle East University	http://www.meu.edu.jo/	28	21	43	27	0	40	89	70
			34	23	44	27	0	40	97	71
15	German Jordanian	http://www.gju.edu.jo/	17	33	11	56	7	81	131	74
			37	27	14	57	6	39	103	77
16	Al Zaytoonah University	http://www.zuj.edu.jo/	6	125	53	108	29	1	132	190
			19	195	47	126	91	1	215	264
17	University of Petra	http://www.uop.edu.jo/	29	20	63	58	0	91	140	121
			?	?	?	?	?	?	?	?
18	Hashemite University	http://www.hu.edu.jo/	51	22	0	55	0	85	158	55
			69	49	0	56	0	34	152	56
19	World Islamic Sciences and Education University	http://www.wise.edu.jo/	45	104	5	39	0	11	160	44
			36	43	14	67	15	45	124	96
20	Irbid National University	http://www.inu.edu.jo/	9	6	13	94	6	148	163	113
			?	?	?	?	?	?	?	?
21	Al Balqa Applied	http://www.bau.edu.jo/	100	36	123	119	11	32	168	253
			115	62	97	74	9	15	192	180
22	Philadelphia University at Jordan	http://www.philadelphia.edu.jo/	1	153	57	44	0	15	169	101
			1	157	59	43	0	13	171	102
23	Princess Sumaya University for Technology	http://www.psut.edu.jo/	2	62	18	24	4	116	180	46
			4	81	21	25	3	101	186	49
24	Al Ahliyya Amman University	http://www.ammanu.edu.jo/	62	77	34	48	0	62	201	82
			43	39	0	38	20	16	98	58
25	Zarqa University	http://www.zu.edu.jo/	31	60	79	84	0	166	257	163
			?	?	?	?	?	?	?	?
26	Al Isra Private University Amman	http://www.iu.edu.jo/	Dn and Err	Dn and Err	Dn and Err	Dn and Err	Dn and Err	Dn and Err	Dn and Err	Dn and Err
			26	279	79	127	16	159	464	222
27	Jadara University	http://www.jadara.edu.jo/	Err	Err	Err	Err	Err	Err	Err	Err
			Err	Err	Err	Err	Err	Err	Err	Err
	Min		*1*	*1*	*0*	*0*	*0*	*0*	*4*	*0*
			1	*1*	*0*	*0*	*0*	*0*	*4*	*0*
	Max		*100*	*153*	*123*	*127*	*29*	*166*	*257*	*253*
			306	*1,273*	*1,389*	*375*	*2,144*	*405*	*1,984*	*3,908*
	Average		*26.04*	*38.2*	*29.04*	*50.32*	*2.96*	*42.96*	*107.2*	*82.32*
			48	*128.105*	*101*	*69.42*	*124*	*51.05*	*227.16*	*294.42*
	Univ no. > aver.		*12*	*8*	*9*	*9*	*8*	*8*	*11*	*8*
			5	*4*	*1*	*5*	*1*	*3*	*2*	*1*

The values presented in Table 2 represent the results of two free online accessibility tools: AccessMonitor (http://www.acessibilidade.gov.pt/accessmonitor/) and the EIII (http://checkers.eiii.eu/). AccessMonitor is a Portuguese free online tool that is used to evaluate the accessibility of different URLs. A scale of 1–10 is used by AccessMonitor tool, where 10 represents the optimum (fully accessible URL), and 0 represents the worst (inaccessible URL). All the AccessMonitor (http://www.acessibilidade.gov.pt/accessmonitor/) results are verified; the results are stable and do not fluctuate within the same session as in the EIII Page Checker. Therefore, Table 2 should be sorted in descending order based on one of the first four columns.

Table 2 Summary of WAVE tool results ranked according to the sum of errors, alerts, and contrast errors (August 28, 2015 and March 1, 2016)

#	*University name*	*University URL*	*AccessMonitor index score WCAG 1.0*	*AccessMonitor index score WCAG 2.0*	*EIII score WCAG 2.0*	*EIII barrier total WCAG 2.0*	*EIII applied test total*
1	University of Jordan	http://www.ju.edu.jo/	4.7	3.9	87	52	1,057
			4.7	3.8	86.38	59	1,163
2	Jordan University of Science and Technology	http://www.just.edu.jo/	4.5	3.8	85	12	462
			4.5	3.8	85.48	12	454
3	Hashemite University	http://www.hu.edu.jo/	2.2	3.0	60	153	581
			2.2	3.0	61.06	162	824
4	Yarmouk University	http://www.yu.edu.jo/	4.9	5.9	92	7	785
			4.3	5.5	99.58	3	767
5	Al Balqa Applied University	http://www.bau.edu.jo/	2.9	3.2	69	176	986
			2.9	3.2	70.31	152	906
6	Mutah University	http://www.mutah.edu.jo/	3.8	4.3	100	19	969
			3.6	4.2	99.57	15	933
7	Zarqa University	http://www.zu.edu.jo/	2.3	3.2	74	36	1,030
			2.5	3.1	73.37	53	1,105
8	Al Ahliyya Amman University	http://www.ammanu.edu.jo/	3.7	3.7	65	110	646
			3.5	3.5	55.56	114	531
9	Princess Sumaya University for Technology	http://www.psut.edu.jo/	6.8	6.1	87	10	783
			6.8	6.1	86.64	10	715
10	American University of Madaba	http://aum.edu.jo/	5.8	6.1	73	138	1,392
			5.8	6.1	82.03	82	1,497
11	Al Al-Bayt University	http://www.aabu.edu.jo/	2.7	3.2	82	19	854
			3.0	3.3	80.86	22	546
12	Applied Science University	http://www.asu.edu.jo/	2.8	2.8	68	68	787
			2.9	3.0	73.95	64	798
13	German Jordanian University	http://www.gju.edu.jo/	5.6	5.4	92	25	982
			5.6	5.4	91.82	25	976
14	Middle East University Jordan	http://www.meu.edu.jo/	4.1	5.1	96	30	928
			4.1	5.1	95.49	35	944
15	University of Petra	http://www.uop.edu.jo/	3.8	4.2	83	44	1,288
			3.3	4.2	83.60	45	1,245
16	World Islamic Sciences and Education University	http://www.wise.edu.jo/	2.5	3.9	82	75	1,251
			4.9	5.4	93.50	35	1,258
17	Philadelphia University at Jordan	http://www.philadelphia.edu.jo/	3.5	4.1	95	8	915
			3.5	4.2	95.19	8	894
18	Amman Arab University	http://www.aau.edu.jo/	3.8	3.7	76	19	370
			3.8	3.7	79.99	18	384

Table 2 Summary of WAVE tool results ranked according to the sum of errors, alerts, and contrast errors (August 28, 2015 and March 1, 2016) (continued)

#	*University name*	*University URL*	*AccessMonitor index score WCAG 1.0*	*AccessMonitor index score WCAG 2.0*	*EIII score WCAG 2.0*	*EIII barrier total WCAG 2.0*	*EIII applied test total*
19	Al Zaytoonah University	http://www.zuj.edu.jo/	5.4	4.2	Problem	Problem	Problem
			4.8	4.1	78.98	36	1,514
20	Jerash Private University	http://www.jpu.edu.jo/	3.7	3.7	75	86	660
			3.2	3.4	74.99	80	579
21	Al Isra Private University Amman	http://www.iu.edu.jo/	7.5	5.5	Down and problem	Down and problem	Down and problem
			5.2	4.3	91.88	32	858
22	Al Hussein bin Talal University	http://www.ahu.edu.jo/	6.1	4.5	84	19	325
			6.1	4.5	83.74	22	333
23	Irbid National University	http://www.inu.edu.jo/	5.8	6.1	75	15	220
			5.8	6.1	75.56	14	245
24	Ajloun National University	http://www.anu.edu.jo/	4.7	4.6	84	21	558
			4.9	4.6	83.61	24	586
25	Arab Academy for Banking and Financial Sciences	http://www.aabfs.org/	5.8	6.1	Problem	Problem	Problem
			5.8	6.1	61.55	64	233
26	Jadara University	http://www.jadara.edu.jo/	6.4	4.6	Problem	Problem	Problem
			4.2	4.0	82.97	118	980
27	Tafila Technical University	http://www.ttu.edu.jo/	3.0	3.7	75.30	60	591
			3.1	3.7	84.64	38	549
	Min		*2.2*	*2.8*	*60*	*7*	*220*
			2.2	*3*	*55.56*	*3*	*233*
	Max		*7.5*	*6.1*	*100*	*176*	*1,392*
			6.8	*6.1*	*99.58*	*162*	*1,514*
	Average		*4.4*	*4.39*	*80.84*	*52.26*	*800.87*
			4.26	*4.35*	*81.94*	*49.70*	*808.04*
	Univ no. >aver.		*13*	*11*	*13*	*8*	*11*
			13	*10*	*16*	*10*	*14*

Noticeably, the results of the EIII are not stable for the website URLs of American University of Madaba, Hashemite University, and Zarqa University. The first accessibility score (WCAG 2.0) of Hashemite University (http://www.hu.edu.jo/) is 61, the number of applied tests is 801, and the number of barriers is 158; moreover, a rerun of the same tool to evaluate the same URL after 2 hours obtains 60, and the number of applied tests is 581 and 153. The results of EIII are also related to http://www.ju.edu.jo/; http://www.just.edu.jo does not change. We could not determine a logical explanation for this deviation in the results related to the URLs shown in Table 2. Therefore, the results presented in Table 2 represent the tool results of the rerun for these three URLs. The highlighted values in Table 2 represent the unstable results of EIII tool. We also noticed that this tool sometimes fails to evaluate a certain URL, such as http://www.bau.edu.jo/, temporarily and shows running errors as well as on a rerun of the tool yield results. The Tafila Technical University website was down on September 12, 2015; however, this website was revisited on September 20, 2015, and was functional; the related measurements are shown in Table 2. The EIII Page Checker failed to process the homepage of Al Isra Private University in Amman on September 20, 2015, because the server was down; on that same date, the website of Al Isra Private University in Amman was functioning, but a rerun of the EIII Page Checker again failed to measure its accessibility score. The EIII Page Checker fails to measure the accessibility score of the following universities: Al Zaytoonah University, Arab Academy for Banking and Financial Sciences, and Jadara University.

Alexa.com is an online analytical tool that collects data about browsing behaviour to 30 million websites using its toolbar; these data form the basis for Alexa's web traffic reporting. The rank of each website within Alexa.com depends mainly on two important factors: number of visitors to this site and the number of web pages they visited within the site under consideration. The popularity of each of the 27 Jordanian university websites is listed in Table 3 according to the Alexa.com listing on August 29, 2015, and

March 1, 2016. Alexa.com fails to identify the rank of four Jordanian universities on August 29, 2015: Ajloun National University, Arab Academy for Banking and Financial Sciences, Jadara University, and Tafila Technical University; nevertheless, the rank of each of these websites is determined globally, as shown in Table 3. Also, Alexa.com fails to identify the rank of three Jordanian universities on March 1, 2016: Irbid National University, Ajloun National University, and Arab Academy for Banking and Financial Sciences; nevertheless, the rank of each of these websites is determined globally on that date.

A list of the top 500 websites in Jordan was obtained from Alexa.com on August 29, 2015, and on March 1, 2016. Data show only eight Jordanian university websites on the list August 29, 2015: University of Jordan rank was 97; Jordan University of Science and Technology: 144; Hashemite University: 150; Yarmouk University: 181; Al Balqa Applied University: 272; Mutah University: 176; Zarqa University: 449; and Al Ahliyya Amman University: 487. Only the rank of University of Jordan enhanced to 89 on March 1, 2016, and all others are declined. New University Websites are listed within the top 500 websites in Jordan on March 1, 2016, like Princess Sumaya University for Technology: 309; German Jordanian University: 324; University of Petra: 361.

Table 3 Summary of Alexa results (August 29, 2015 and March 1, 2016)

#	*University name*	*University URL*	*Rank in Jordan (Alexa)*	*Global rank (Alexa)*
1	University of Jordan	http://www.ju.edu.jo/	97	57,997
			89	53,824
2	Jordan University of Science and Technology	http://www.just.edu.jo/	144	144,520
			258	122,045
3	Hashemite University	http://www.hu.edu.jo/	150	126,550
			229	151,547
4	Yarmouk University	http://www.yu.edu.jo/	181	137,911
			336	157,920
5	Al Balqa Applied University	http://www.bau.edu.jo/	272	379,178
			642	341,437
6	Mutah University	http://www.mutah.edu.jo/	276	337,333
			978	488,719
7	Zarqa University	http://www.zu.edu.jo/	449	641,217
			1,861	734,601
8	Al Ahliyya Amman University	http://www.ammanu.edu.jo/	487	51,810
			23,698	58,109
9	Princess Sumaya University for Technology	http://www.psut.edu.jo/	556	459,176
			309	348,378
10	American University of Madaba	http://aum.edu.jo/	714	827,896
			1,133	780,344
11	Al Al-Bayt University	http://www.aabu.edu.jo/	728	464,828
			898	310,516
12	Applied Science University	http://www.asu.edu.jo/	852	917,195
			1,580	906,741
13	German Jordanian University	http://www.gju.edu.jo/	858	623,257
			324	290,566
14	Middle East University Jordan	http://www.meu.edu.jo/	1,275	518,792
			2,915	388,290
15	University of Petra	http://www.uop.edu.jo/	1,401	232,296
			361	163,460
16	World Islamic Sciences and Education University	http://www.wise.edu.jo/	1,417	583,174
			996	572,980
17	Philadelphia University at Jordan	http://www.philadelphia.edu.jo/	1,825	228,685
			1,297	213,420
18	Amman Arab University	http://www.aau.edu.jo/	1,950	1,903,311
			1,535	1,351,504
19	Al Zaytoonah University	http://www.zuj.edu.jo/	2,102	743,265
			666	434,676
20	Jerash Private University	http://www.jpu.edu.jo/	2,471	1,875,779
			1,442	963,169
21	Al Isra Private University Amman	http://www.iu.edu.jo/	2,511	1,388,837
			1,655	934,296

Table 3 Summary of Alexa results (August 29, 2015 and March 1, 2016) (continued)

#	*University name*	*University URL*	*Rank in Jordan (Alexa)*	*Global rank (Alexa)*
22	Al Hussein bin Talal University	http://www.ahu.edu.jo/	2,684	848,641
			1.010	679.311
23	Irbid National University	http://www.inu.edu.jo/	3,139	3,023,206
			NA	3,951,056
24	Ajloun National University	http://www.anu.edu.jo/	NA	3,010,386
			NA	7,916,189
25	Arab Academy for Banking and Financial Sciences	http://www.aabfs.org/	NA	1,441,126
			NA	646,835
26	Jadara University	http://www.jadara.edu.jo/	NA	2,280,191
			2197	1,112,398
27	Tafila Technical University	http://www.ttu.edu.jo/	NA	2,734,973
			3,718	1,176,644
	Min		*97*	*51,810*
			89	*53.824*
	Max		*3,139*	*3,023,206*
			23,698	*7,916,189*
	Average		*1,153.87*	*962,278.89*
			2,088.63	*935,147.22*
	Univ no. >aver.		*10*	*8*
			4	*6*

Table 4 Summary of QS university rankings of Jordanian universities (September 21, 2015)

#	*University name*	*Arab region ranking*	*QS score*	*Type*
1	University of Jordan	8	79.6	Public
2	Jordan University of Science and Technology	10	74.2	Public
3	Yarmouk University	29	51.2	Public
4	University of Petra	44	36.5	Private
5	Princess Sumaya University for Technology	47	34.1	Private
6	Hashemite University	49	31.9	Public
7	Applied Science University	82	NA	Private
8	Philadelphia University at Jordan	86	NA	Private
9	German Jordanian University	91	NA	Public

The Times Higher Education World University Rankings 2014–2015 list does not include any university in the Arab region; thus, it cannot be used in this study. The academic ranking of world universities listed the top 500 world universities, which also does not include any Jordanian university in their 2015 ranking; hence, it is not used in this study. The QS university rankings of the Arab Region in 2015 include only 9 out of the 27 Jordanian universities in the list of the top 100 Arab Universities, as shown in Table 4. Therefore, this ranking could not be considered in this study because the other Jordanian universities are not included. Some ranks of Jordanian universities are contradicting in Tables 4 and 5 because of the different algorithms adopted by different university rankings. The results in Table 4 indicate five public Jordanian universities and four private Jordanian universities in the list of the top 100 Arab universities. The population of Jordan constitutes approximately 2% of the Arab world (Wikipedia (2016), but the top Jordanian universities constitute 9% of the top 100 Arab universities. This result indicates that the quality of higher education in Jordan exceeds its population percentage in the Arab world. Table 4 includes one measurement for each of the parameters since the values of these parameters have changed since September 21, 2015, till March 1st, 2016.

The only comprehensive university rankings that include all Jordanian universities is webometrics ranking of world universities published by Cybermetrics Lab and is thus used in this study. Table 5 shows the university rankings-related information collected from webometrics ranking of world universities. Jordanian universities are ranked in Table 5 according to their country rank on August 29, 2015. The country rank column in Table 5 has numbers that exceed 27, and this is due to the presence of colleges and institutes besides Jordanian universities in the webometrics list of Jordan (http://www.webometrics.info/en/aw/Jordan) like Al Quds College, Institute of Banking Studies, Jordan Media Institute, and Queen Noor Civil Aviation Technical College.

Table 5 Summary of Webometrics Rankings of Jordanian universities (August 29, 2015 and March 1, 2016)

#	*University name*	*World ranking*	*Continental ranking*	*Country rank*	*Presence*	*Impact rank*	*Openness*	*Excellence*
1	University of Jordan	1,620	14	1	896	1,334	99,999	1,270
		1,010	7	1	536	1,662	309	1,276
2	Jordan University of Science and Technology	1,979	21	2	3,957	5,506	2,269	1,433
		1,907	23	2	3,726	5,161	2,111	1,439
3	Yarmouk University	2,075	26	3	1,256	5,501	1,814	1,878
		1,969	25	3	841	4,901	1,779	1,882
4	Hashemite University	2,460	44	4	6,293	6,830	3,269	1,616
		2,292	38	5	6,004	6,320	3,005	1,622
5	Al Balqa Applied University	3,058	66	5	7,418	7,952	5,226	2,133
		2,742	57	6	4,327	7,604	4,837	2,138
6	Philadelphia University at Jordan	3,060	67	6	1,282	2,282	99,999	3,093
		2,056	28	4	794	2,248	1,202	3,092
7	Princess Sumaya University for Technology	3,201	75	7	4,686	3,848	9,315	3,482
		3,295	80	8	5,271	4,287	9,959	3,481
8	Al Al-Bayt University	3,363	82	8	3,391	8,918	3,385	2,717
		3,119	71	7	3,061	8,491	3,425	2,717
9	Mutah University	3,740	96	9	9,534	8,806	6,626	2,554
		3,381	87	9	9,941	8,252	5,256	2,554
10	University of Petra	4,317	110	10	6,951	3,314	1,177	5,490
		4,018	104	10	5,019	3,225	1,053	5,484
11	Middle East University Jordan	4,372	111	11	10,694	4,746	4,782	4,318
		4,645	121	13	10,261	6,042	4,683	4,315
12	Applied Science University	4,496	117	12	6,864	9,926	8,728	2,983
		4,045	106	11	6,448	9,554	8,413	2,982
13	German Jordanian University	4,970	126	13	12,659	8,022	6,109	3,656
		4,530	117	12	12,400	7,506	6,149	3,654
14	Al Hussein bin Talal University	5,417	140	14	11,605	11,362	9,872	2,881
		4,832	128	15	11,180	11,101	9,093	2,881
15	Al Zaytoonah University	5,887	151	15	6,921	13,938	4,499	3,048
		5,573	144	16	6,985	14,089	4,227	3,047
16	Tafila Technical University	7,609	179	16	14,143	13,297	15,085	2,983
		7,191	166	17	15,022	12,866	15,866	2,982
17	Al Ahliyya Amman University	7,963	186	17	710	13,122	7,160	4,569
		4,806	127	14	363	8,024	6,171	4,565
18	Zarqa University	8,344	195	18	9,531	15,450	8,350	3,364
		7,449	174	18	9,813	15,015	5,995	3,363
19	Arab Academy for Banking and Financial Sciences	9,945	234	19	13,847	11,911	11,115	4,569
		9,851	225	19	14,231	11,972	11,482	4,565
20	Amman Arab University	11,567	272	20	11,226	14,165	16,600	4,318
		10,627	239	20	11,462	13,850	13,839	4,315
21	Jerash Private University	11,928	281	21	18,240	15,432	16,277	3,750
		11,824	260	21	18,696	15,417	16,200	3,746
22	American University of Madaba	12,315	292	22	852	16,566	13,656	4,898
		12,585	278	23	867	16,682	13,669	4,892
23	World Islamic Sciences and Education University	12,327	293	23	4,621	15,239	18,271	4,569
		12,367	274	22	5,772	15,494	17,386	4,565
24	Irbid National University	16,823	512	27	18,722	14,397	20,114	5,490
		15,744	423	26	16,668	14,418	18,653	5,484
25	Al Isra Private University Amman	17,214	529	28	11,947	18,863	11,070	5,490
		17,409	493	29	11,194	18,866	12,023	5,484
26	Jadara University	18,969	658	32	13,899	21,657	13,947	4,569
		13,753	330	24	16,552	16,359	12,477	4,565
27	Ajloun National University	19,176	670	34	15,725	20,304	12,576	5,490
		17,591	502	30	15,383	18,583	11,779	5,484

Table 6 presents a summary of all ranks already presented in Tables 1, 2, 3, and 5; these ranks are used in this study to determine whether any correlation exists between two variables (ranks). Correlation coefficients help in obtaining precise measurements about the strengths, magnitudes, and directions of the relationship between any two of these ranks under study. Spearman's rank correlation is used to determine the relationship between two ranked variables understudy. Some values presented in Table 6 are ignored during the computations of correlation coefficients.

IBM SPSS version 17 is used in this study to compute the Kendall's tau_b correlation coefficients. Table 1 shows that the WAVE tool failed on March 1, 2016, to evaluate 10 out of the 27 Jordanian university websites. Therefore, the results of Wave tool on March 1, 2016, are not included in the computation of their rank correlations with the rank of other metrics. The ranked-based correlation analysis is used to discover the concordance of 18 metrics. The 18 metrics include the 8 metrics presented in table 1 for the date August 28, 2015. The other ten metrics represent measurements conducted on August 28, 2015, and March 1, 2016, therefore, each of the cells of the following metrics has two measurements on different dates: first three metrics presented in Table 2, first metric presented in Table 3, third metric presented in Table 5. SPSS Kendall's tau_b correlation coefficients of these 18 metrics show a significant (positive/negative) agreement between 36 pairs. Table 7 shows only the significant concordant pairs ordered in descending order. The top 9 concordances presented in Table 7 are between metrics of the same tool, but afterward, you can witness positive agreements between Alexa and Webometrics. Furthermore, Table 7 presents agreements between (AccessMonitor-1.0 and EIII-Score-2.0), (AccessMonitor-2.0 and EIII-Score-2.0), (WAVE's HTML5 Rank and AccessMonitor-1.0), and (WAVE Errors Rank and EIII-Score-2.0).

Table 6 Summary of web accessibility, web traffic, and university ranks

I	*University name*	*WAVE's errors sum*	*WCAG 1.0 AccessMonitor*	*WCAG 2.0 AccessMonitor*	*WCAG 2.0 EIII Score*	*Alexa rank*	*Webometrics rank*
1	Ajloun National University	9	12	9	10	NA	27
2	Al Ahliyya Amman University	24	18	19	22	8	17
3	Al Al-Bayt University	10	24	25	12	11	8
4	Al Balqa Applied University	21	22	23	20	5	5
5	Al Hussein bin Talal University	4	4	11	9	22	14
6	Al Isra Private University Amman	NA	1	6	NA	21	25
7	Al Zaytoonah University	16	9	14	NA	19	15
8	American University of Madaba	1	5	2	19	10	22
9	Amman Arab University	5	17	20	14	18	20
10	Applied Science University	13	23	27	21	12	12
11	Arab Academy for Banking and Financial Sciences	7	7	4	NA	NA	19
12	German Jordanian University	15	8	7	5	13	13
13	Hashemite University	18	27	26	23	3	4
14	Irbid National University	20	6	3	17	23	24
15	Jadara University	NA	3	10	NA	NA	26
16	Jerash Private University	11	19	21	16	20	21
17	Jordan University of Science and Technology	3	13	18	8	2	2
18	Middle East University Jordan	14	14	8	2	14	11
19	Mutah University	6	15	12	1	6	9
20	Philadelphia University at Jordan	22	20	15	3	17	6
21	Princess Sumaya University for Technology	23	2	1	7	9	7
22	Tafila Technical University	12	21	22	15	NA	16
23	University of Jordan	8	11	16	6	1	1
24	University of Petra	17	16	13	11	15	10
25	World Islamic Sciences and Education University	19	25	17	13	16	23
26	Yarmouk University	2	10	5	4	4	3
27	Zarqa University	25	26	24	18	7	18

Table 7 Kendall's tau_b significant correlation coefficients at the 0.01 level and the 0.05 level

I	*Pair*	*Value*
1	τ (Webometrics-Aug-2015-Rank, Webometrics-March-2016-Rank)	1.000**
2	τ (AccessMonitor-2.0-Aug-2015-Rank, AccessMonitor-2.0-March-2016-Rank)	0.858**
3	τ (EIII-Score-2.0-Aug-2015-Rank, EIII-Score-2.0-March-2016-Rank)	0.794**
4	τ (AccessMonitor-1.0-Aug-2015-Rank, AccessMonitor-1.0-March-2016-Rank)	0.744**
5	τ (WAVE's Structural Element Rank, WAVE's Feature Sum Rank)	0.733**
6	τ (AccessMonitor-1.0-March-2016-Rank, AccessMonitor-2.0-March-2016-Rank)	0.709**
7	τ (AccessMonitor-2.0-Aug-2015-Rank, AccessMonitor-1.0-March-2016-Rank)	0.704**
8	τ (AccessMonitor-1.0-Aug-2015-Rank, AccessMonitor-2.0-Aug-2015-Rank)	0.687**
9	τ (WAVE's Feature Rank, WAVE's Features Sum Rank)	0.647**
10	τ (Alexa-Aug-2015-Rank, Webometrics-Aug-2015-Rank)	0.565**
11	τ (Alexa-Aug-2015-Rank, Webometrics-March-2016-Rank)	0.565**
12	τ (AccessMonitor-1.0-Aug-2015-Rank, EIII-Score-2.0-Aug-2015-Rank)	0.556**
13	τ (Webometrics-Aug-2015-Rank, Alexa-March-2016-Rank)	0.507**
14	τ (Alexa-March-2016-Rank, Webometrics-March-2016-Rank)	0.507**
15	τ (WAVE's Contrast Errors Rank, WAVE's Error Sum Rank)	0.487**
16	τ (WAVE's Alert Rank, WAVE's Errors Sum Rank)	0.473**
17	τ (AccessMonitor-2.0-Aug-2015-Rank, EIII-Score-2.0-Aug-2015-Rank)	0.439**
18	τ (WAVE's HTML5 Rank, AccessMonitor-1.0-Aug-2015-Rank)	0.420**
19	τ (WAVE Errors Rank, EIII-Score-2.0-Aug-2015-Rank)	0.399**
20	τ (WAVE Errors Rank, EIII-Score-2.0-March-2016-Rank)	0.387**
21	τ (Alexa-Aug-2015-Rank, Alexa-March-2016-Rank)	0.385*
22	τ (EIII-Score-2.0-Aug-2015-Rank, AccessMonitor-2.0-March-2016-Rank)	0.383*
23	τ (WAVE's Feature Rank, WAVE's Structural Elements Rank)	0.380**
24	τ (WAVE Errors Rank, AccessMonitor-2.0-Aug-2015-Rank)	0.367*
25	τ (WAVE Errors Rank, AccessMonitor-1.0-Aug-2015-Rank)	0.360*
26	τ (AccessMonitor-2.0-March-2016-Rank, EIII-Score-2.0-March-2016-Rank)	0.356**
27	τ (WAVE's Errors Sum Rank, WAVE's Features Sum Rank)	-0.347-*
28	τ (WAVE's HTML5 Rank, WAVE's Errors Sum Rank)	0.333*
29	τ (WAVE's Structural Element Rank, WAVE's Error Sum Rank)	-0.320-*
30	τ (WAVE Errors Rank, AccessMonitor-1.0-March-2016-Rank)	0.320*
31	τ (AccessMonitor-1.0-Aug-2015-Rank, EIII-Score-2.0-Aug-2015-Rank)	0.320*
32	τ (AccessMonitor-2.0-Aug-2015-Rank, EIII-Score-2.0-March-2016-Rank)	0.316*
33	τ (WAVE Errors Rank, AccessMonitor-2.0-March-2016-Rank)	0.313*
34	τ (WAVE's HTML5 Rank, AccessMonitor-1.0-March-2016-Rank)	0.313*
35	τ (WAVE's Errors Sum Rank, AccessMonitor-1.0-Aug-2015-Rank)	0.300*
36	τ (WAVE's Alert Rank, AccessMonitor-1.0-Aug-2015-Rank)	0.293*

Notes: **Correlation is significant at the 0.01 level (2-tailed).
*Correlation is significant at the 0.05 level (2-tailed).

Furthermore, this study aims to explore whether a difference exists between the accessibility of public and private Jordanian universities. The numbers of private and public universities in this study are 16 and 11, respectively. A non-parametric independent sample t-test on ranked data called Mann-Whitney U test is used in this study at α significance level, which equals 0.05. Mann-Whitney U test is used to examine whether two groups of interest (private and public universities) come from the same distribution for a small sample. The AccessMonitor index score (WCAG 1.0), AccessMonitor index score (WCAG 2.0), and EIII score (WCAG 2.0) accessibility values presented in Table 8 are used to conduct a statistical analysis to determine

whether public and private Jordanian university websites differ in their compliance with accessibility guidelines.

First, the H_0 (null) and H_1 (alternative) hypotheses of AccessMonitor index score (WCAG 1.0) should be defined.

H_0 No difference exists between the ranks of private and public AccessMonitor index scores (WCAG 1.0).

H_1 A difference exists between the ranks of private and public AccessMonitor index scores (WCAG 1.0).

The statistical analysis using SPSS shows that the null hypothesis (H_0) should be accepted because the Mann-Whitney U test obtains 0.178, which is not statistically significant at ($\alpha = 0.05$).

Second, the H_0 (null) and H_1 (alternative) hypotheses of AccessMonitor index score (WCAG 1.0) should be defined.

H_0 No difference exists between the ranks of private and public AccessMonitor index scores (WCAG 2.0).

H_1 A difference exists between the ranks of private and public AccessMonitor index scores (WCAG 2.0).

The statistical analysis using SPSS shows that the null hypothesis (H_0 should be accepted because the Mann-Whitney U test obtains 0.195, which is not statistically significant at ($\alpha = 0.05$).

Third, H_0 (null) and H_1 (alternative) hypotheses of the EIII score (WCAG 2.0) should be defined.

H_0 No difference exists between the ranks of private and public EIII scores (WCAG 2.0).

H_1 A difference exists between the ranks of private and public EIII scores (WCAG 2.0).

The statistical analysis using SPSS shows that the null hypothesis (H_0) should be accepted because the Mann-Whitney U test obtains 0.379, which is not statistically significant at ($\alpha = 0.05$).

Table 8 Summary of accessibility metrics

I	*University name*	*University type*	*AccessMonitor index score WCAG 1.0*	*AccessMonitor index score WCAG 2.0*	*EIII score WCAG 2.0*
1	Ajloun National University	Private	4.7	4.6	84
2	Al Ahliyya Amman University	Private	3.7	3.7	65
3	Al Isra Private University Amman	Private	7.5	5.5	Down and problem
4	Al Zaytoonah University	Private	5.4	4.2	Problem
9	American University of Madaba	Private	5.8	6.1	73
6	Amman Arab University	Private	3.8	3.7	76
7	Applied Science University	Private	2.8	2.8	68
8	Arab Academy for Banking and Financial Sciences	Private	5.8	6.1	Problem
9	Irbid National University	Private	5.8	6.1	75
10	Jadara University	Private	6.4	4.6	Problem
11	Jerash Private University	Private	3.7	3.7	75
12	Middle East University Jordan	Private	4.1	5.1	96
13	Philadelphia University at Jordan	Private	3.5	4.1	95
14	University of Petra	Private	3.8	4.2	83
15	Zarqa University	Private	2.3	3.2	74
16	Princess Sumaya University for Technology	Private non-profit university	6.8	6.1	87
17	Al Al-Bayt University	Public	2.7	3.2	82
18	Al Balqa Applied University	Public	2.9	3.2	69
19	Al Hussein bin Talal University	Public	6.1	4.5	84
20	German Jordanian University	Public	5.6	5.4	92
21	Hashemite University	Public	2.2	3	60
22	Jordan University of Science and Technology	Public	4.5	3.8	85
23	Mutah University	Public	3.8	4.3	100
24	Tafila Technical University	Public	3	3.7	75.3
25	University of Jordan	Public	4.7	3.9	87
26	World Islamic Sciences and Education University	Public	2.5	3.9	82
27	Yarmouk University	Public	4.9	5.9	92

5 Conclusions and future work

This study uses a number of free online tools to assess the accessibility of Jordanian university websites against W3C's WCAG standard. Alexa.com. Subsequently, this study aims to rediscover the relationships between accessibility, web traffic, and university rankings. A fairly strong positive correlation is observed between different accessibility metrics, whereas a fairly weak negative correlation is observed between one of the accessibility metrics and Alexa Ranks; a fairly strong positive correlation is also observed between Alexa Ranks and Webometrics university rankings. Finally, this study explores the difference between the accessibility of public and private Jordanian universities; the statistical analysis using SPSS Mann-Whitney U test shows no difference between the accessibility of private and public Jordanian universities. Future work will explore the use of more metrics and a larger number of universities to assess the relationships between them.

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